

SARAJEVO-BJELAVE, Bosnia and Herzegovina

WMO: 146540

Lat: 43.8678N

Lon: 18.4228E

Elev: 2096

StdP: 13.62

Time Zone: 1.00 (EUC)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.8	16.5	4.3	7.4	14.1	9.2	9.5	19.0	14.5	44.2	13.0	40.3	3.1	110	0.451

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.6	91.8	67.6	88.3	66.9	84.9	66.0	71.0	85.4	68.9	83.2	67.1	81.1	4.7	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.1	103.9	79.7	63.8	95.9	75.8	62.2	90.5	73.0	36.2	85.7	34.3	83.1	32.8	81.3	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.7	11.0	9.5	DB	6.2	96.9	6.4	2.9	1.6	99.0	-2.1	100.7	-5.7	102.4	-10.4	104.5
			WB	5.4	73.7	6.0	2.8	1.0	75.7	-2.5	77.3	-5.9	78.9	-10.3	81.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.2	32.8	35.9	43.6	52.7	59.3	65.9	70.2	70.6	61.6	53.7	44.3	34.4	(6)
(7)		DBStd	15.38	9.41	9.64	8.89	7.07	6.82	7.21	6.31	6.42	7.22	7.56	9.03	8.75	(7)
(8)		HDD50	1987	534	398	231	52	10	2	0	1	5	52	213	489	(8)
(9)		HDD65	5237	997	813	663	373	198	75	25	24	146	353	621	949	(9)
(10)		CDD50	2779	2	5	33	132	298	479	625	638	354	166	42	5	(10)
(11)		CDD65	555	0	0	0	3	21	102	185	197	45	2	0	0	(11)
(12)		CDH74	5346	0	0	1	57	253	1003	1798	1805	399	30	0	0	(12)
(13)		CDH80	2024	0	0	0	5	46	325	748	772	128	0	0	0	(13)
(14)	Wind	WSAvg	3.8	3.3	3.8	4.4	4.3	3.8	3.8	3.9	3.7	3.5	3.9	3.4	(14)	
(15)	Precipitation	PrecAvg	36.2	2.7	2.6	2.6	3.0	3.0	3.4	3.0	2.6	3.1	3.2	3.6	3.6	(15)
(16)		PrecMax	49.3	6.3	7.2	6.9	5.5	8.0	6.3	7.5	6.2	10.2	10.2	7.0	11.6	(16)
(17)		PrecMin	24.7	0.2	0.5	0.0	0.3	1.1	0.9	0.3	0.5	0.2	0.0	0.7	0.0	(17)
(18)		PrecStd	6.0	1.5	1.6	1.4	1.1	1.5	1.5	1.5	1.4	1.9	2.0	1.7	2.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.3	61.3	71.2	80.2	84.9	91.2	95.7	97.0	90.1	77.5	68.1	58.9	(19)
(20)		MCWB	48.3	48.3	50.6	56.7	63.8	68.9	68.8	68.4	65.1	59.7	56.8	51.9	(20)	
(21)		2%	DB	53.5	56.1	65.4	74.1	80.2	87.2	92.0	92.0	83.3	73.0	63.9	55.3	(21)
(22)		MCWB	46.1	45.9	49.0	55.2	61.2	67.0	67.8	67.5	64.0	58.6	53.6	48.0	(22)	
(23)		5%	DB	49.7	53.0	61.1	69.3	76.3	83.8	88.1	88.2	78.0	69.1	61.0	52.1	(23)
(24)		MCWB	43.8	44.7	47.8	53.2	59.3	65.4	67.2	66.7	61.7	56.9	51.7	45.6	(24)	
(25)		10%	DB	45.9	49.8	57.4	65.0	72.5	80.2	84.1	84.2	73.7	65.3	57.5	48.2	(25)
(26)		MCWB	41.0	42.8	46.5	51.3	58.1	63.9	66.4	65.9	60.2	55.4	49.7	42.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.9	49.8	52.6	59.9	66.8	73.0	74.8	74.6	69.6	62.1	58.2	52.4	(27)
(28)		MCDWB	55.2	59.2	64.6	74.8	81.0	85.8	91.0	91.4	82.7	73.2	66.8	58.3	(28)	
(29)		2%	WB	46.5	47.0	50.5	56.8	63.3	69.1	70.7	70.9	65.0	59.6	54.4	48.7	(29)
(30)		MCDWB	52.8	54.8	62.3	70.4	75.5	82.9	85.8	85.1	77.8	69.1	62.3	54.5	(30)	
(31)		5%	WB	44.0	45.1	48.8	54.6	61.0	67.0	68.7	68.4	63.0	58.0	52.3	46.1	(31)
(32)		MCDWB	49.4	52.2	59.2	66.2	72.2	80.5	83.7	82.5	75.1	66.8	60.1	51.6	(32)	
(33)		10%	WB	41.3	43.2	47.1	52.6	59.0	64.9	67.0	66.6	61.2	56.3	50.1	43.3	(33)
(34)		MCDWB	45.5	49.5	56.7	62.6	70.0	77.4	81.1	80.7	71.7	64.0	56.8	47.9	(34)	
(35)	Mean Daily Temperature Range		MDBR	11.0	13.2	16.7	18.9	20.1	20.7	22.6	23.8	19.6	18.0	13.4	9.6	(35)
(36)		5% DB	MCDBR	14.4	18.4	24.9	27.4	27.1	26.9	29.4	30.4	27.3	24.4	17.7	12.4	(36)
(37)		MCWBR	9.1	10.8	12.4	12.5	11.5	10.6	11.1	11.4	11.7	12.0	10.3	8.8	(37)	
(38)		5% WB	MCDBR	13.5	16.1	20.7	23.6	23.6	24.7	26.3	26.2	23.6	21.5	15.5	11.7	(38)
(39)		MCWBR	8.9	9.8	10.8	11.8	11.4	11.2	11.7	11.9	11.1	11.4	9.6	8.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.311	0.341	0.385	0.438	0.434	0.426	0.427	0.428	0.404	0.370	0.333	0.311	(40)	
(41)		taud	2.394	2.282	2.217	2.105	2.167	2.240	2.237	2.233	2.279	2.369	2.421	2.413	(41)	
(42)		Ebn at Noon	257	266	268	262	266	269	267	262	259	253	246	245	(42)	
(43)		Edh at Noon	24	32	39	48	46	43	43	42	37	30	24	22	(43)	
(44)	All-Sky Solar Radiation	RadAvg	454	677	1049	1342	1660	1902	1971	1776	1239	861	526	386	(44)	
(45)		RadStd	65	82	140	175	150	165	153	169	125	98	63	68	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (31 neighbors)	+0.99	N/A	N/A	N/A	N/A	N/A	-191	+202	N/A	N/A	(47)

Nomenclature: See separate page